

AI-Powered FAQ Chatbot

Internship Project Report • Elevate Labs • Web Development Domain

1. Introduction

The AI-Powered FAQ Chatbot is a full-stack web application developed as part of the Elevate Labs Internship Project Phase. The project addresses a common business need: providing instant, accurate answers to frequently asked questions without requiring human support agents. Modern users expect immediate responses, and this chatbot delivers that experience by combining a fast local keyword-matching engine with a cloud-based AI model as a fallback.

The system is built entirely with open and widely-adopted technologies — React on the frontend and Node.js/Express on the backend — making it easy to extend, customise, and deploy to any cloud provider.

2. Abstract

This project implements a two-layer intelligent answer engine. The first layer uses a custom keyword-scoring algorithm that scans the user's input against a structured FAQ knowledge base and returns an instant match when keyword overlap is detected. The second layer invokes the Claude AI API (claude-haiku) when no confident FAQ match is found, enabling the bot to handle open-ended or novel queries gracefully.

The frontend delivers a real-time chat interface with a browseable FAQ panel, category filters, per-message confidence indicators, and a thumbs-up/thumbs-down feedback system. Negative feedback and unmatched queries are logged server-side for future knowledge base improvement — closing the loop between user behaviour and system quality.

3. Tools Used

Layer	Technology	Purpose
Frontend	React 18	UI framework for the chat interface
Frontend	CSS Variables / Custom CSS	Theming, dark UI, responsive layout
Backend	Node.js + Express.js	REST API server
Backend	CORS + dotenv	Cross-origin requests, env config
AI	Anthropic Claude (Haiku)	AI fallback for unmatched questions
Algorithm	Keyword Scoring Engine	Multi-word keyword match & ranking
Storage	In-memory array	Unanswered query logging (server-side)

4. Steps Involved in Building the Project

Step 1 — Knowledge Base Design: Defined 12 FAQ entries across 5 categories (Account, Billing, Technical, Features, Support) with multi-word keyword arrays for flexible matching.

Step 2 — Keyword Scoring Engine: Built a scoring algorithm in Express that tokenises the user message, matches it against all FAQ keyword arrays, and returns the best-scoring entry above a minimum threshold.

Step 3 — AI Fallback Integration: When the keyword score is below threshold, the server calls the Anthropic API with the FAQ knowledge base injected into the system prompt plus the last 6 messages of conversation history for context.

Step 4 — REST API Development: Created four endpoints — GET /api/faqs (browse), POST /api/chat (send message), POST /api/feedback (thumbs rating), GET /api/admin/unanswered (improvement log).

Step 5 — React Chat Interface: Designed a dark-themed chat UI with animated message bubbles, a typing indicator, source badges (FAQ Match / AI Answer / Fallback), and confidence progress bars.

Step 6 — FAQ Sidebar Panel: Built a collapsible sidebar with live search, category filter tabs, and colour-coded dot indicators. Clicking any FAQ entry sends it as a user message.

Step 7 — Feedback Loop: Each bot response includes thumbs-up/thumbs-down buttons. Negative feedback appends the query to the unanswered log for human review.

Step 8 — Testing & Polish: Tested edge cases (empty input, API failure, multi-turn context), added responsive CSS for mobile, and wrote a full README with deployment instructions.

5. Conclusion

The AI-Powered FAQ Chatbot successfully demonstrates the integration of traditional rule-based NLP (keyword matching) with modern generative AI to create a robust, production-ready support assistant. The two-layer architecture ensures low latency for common queries while maintaining broad coverage for complex or unexpected inputs.

The project reinforced key full-stack skills: RESTful API design, React component architecture, real-time UI patterns, and responsible AI integration with proper fallback handling. Future enhancements could include a MongoDB-backed persistent FAQ editor, admin analytics dashboard, and Retrieval-Augmented Generation (RAG) for more accurate AI answers.

Project	AI-Powered FAQ Chatbot
Domain	Web Development
Program	Elevate Labs Internship — Project Phase (2 Weeks)
Tech Stack	React · Node.js · Express · Anthropic Claude API